

# Yonghoon Jeong

Lecturer (Assistant Professor-track), Department of Cyber Science  
Researcher, Institute of Naval Research  
Republic of Korea Naval Academy

yhjeong@navy.ac.kr | +82 10-8834-3715 | [Google Scholar](#) | [Homepage](#)

## Research Interests

---

Wireless Communication Systems  
Physical-Layer Wireless Sensing and Networking  
Low-Power Wide-Area Networks (LPWANs)  
Ocean Internet of Things (IoT)  
Satellite Communications

## Education

---

**M.S. in Electrical and Computer Engineering** 2024 – 2026  
- Seoul National University, Seoul, South Korea  
- Advisor: Prof. Saewoong Bahk  
**B.S. in Electronics and Communications Engineering** 2018 – 2023  
- Kwangwoon University, Seoul, South Korea

## Professional Experience

---

**Republic of Korea Naval Academy (ROKNA)**, Changwon, South Korea 2026 – Present  
- Lecturer (Assistant Professor-track), Department of Cyber Science  
- Researcher, Institute of Naval Research  
**Seoul National University (SNU)**, Seoul, South Korea 2024 – 2026  
- Researcher, Network Lab (NETLAB), Institute of New Media and Communications (INMC)  
**Electronics and Telecommunications Research Institute (ETRI)**, Seongnam, South Korea 2023  
- Researcher, AI-SoC Infrastructure Research Section  
**Korea Advanced Institute of Science and Technology (KAIST)**, Daejeon, South Korea 2022  
- Research Intern, Unmanned Systems Research Group (USRG), School of Electrical Engineering (EE)  
- Advisor: Prof. David Hyunchul Shim

## Publications

---

### International Papers

- “Receiver-Aware Chirp Waveform Optimization for Mobile Underwater LoRa Communications,” in preparation.
- “On the Symbol Error Rate of Non-Linear Chirp Modulation for LoRa Under Asynchronous Interference,” under review, *IEEE Wireless Communications Letters*. (SCIE, IF: 5.5)
- **Yonghoon Jeong**, Youngwook Son, and Saewoong Bahk, “On the Interference Resilience of Non-Linear Chirp Modulation for LoRa: An Analytical Insight into Energy Scattering Effect,” *IEEE Communications Letters*, vol. 30, pp. 2238–2242, Jun. 2026. (SCIE, IF: 4.5) [Link](#)
- Dongrack Choi, Yubin Choi, **Yonghoon Jeong**, Jeongyeup Paek, and Saewoong Bahk, “D<sup>2</sup>-PLC: Holistic Design and Implementation of High-Data-Rate Duplex DC Power Line Communication Network,” *IEEE Internet of Things Journal*, vol. 12, pp. 45325–45340, Nov. 2025. (SCIE, IF: 8.9) [Link](#)
- Dongrak Choi, **Yonghoon Jeong**, Yubin Choi, and Saewoong Bahk, “A Programmable High-Throughput Duplex DC-PLC Testbed for Power and Data Integration,” *IEEE ICNP 2025*, Seoul, Korea, Oct. 19–22, 2025. [Link](#)
- **Yonghoon Jeong** and Youngseok Choi, “Diffusion-Phys: Noise-Robust Heart Rate Estimation from Facial Videos

via Diffusion Models,” *Biomedical Engineering Letters*, vol. 15, pp. 575–585, Apr. 2025. (SCIE, IF: 3.4) [Link](#)

## Domestic Papers

- **Yonghoon Jeong** and Saewoong Bahk, “A Study on Direct Current Power Line Communication (DC-PLC) with Applications in Automotive, Maritime, and Aviation,” *The Conference of the Korea Institute of Communications and Information Sciences (KICS)*, Gyeongju, Korea, Nov. 20–22, 2024.
- **Yonghoon Jeong**, “Development of Non-contact Heart Rate Measurement Technology Based on Facial Video,” *The Conference of the Korean Institute of Information Scientists and Engineers (KIISE)*, Jeju, Korea, Jun. 23–25, 2021.

## Awards & Scholarships

|                                                                |           |
|----------------------------------------------------------------|-----------|
| <b>Scholarship Recipient</b>                                   | 2024–2025 |
| - Brain Korea 21 (BK21), National Research Foundation of Korea |           |
| <b>Scholarship Recipient</b>                                   | 2022–2023 |
| - Korea Scholarship Foundation for the Future Leaders (KOSFFL) |           |
| <b>Excellent Paper Awards</b>                                  | 2021      |
| - Korea Computer Conference 2021                               |           |
| <b>University President’s Award</b>                            | 2020      |
| - Seoul Capstone Design, Seoul                                 |           |
| <b>Scholarship Recipient</b>                                   | 2020      |
| - Samsung Scholarship Foundation                               |           |
| <b>Dean’s List</b>                                             | 2020      |
| - Highest GPA                                                  |           |

## Teaching Experience

|                                                                            |      |
|----------------------------------------------------------------------------|------|
| <b>Creative Engineering Design</b> , Prof. Saewoong Bahk                   | 2024 |
| <b>Electronic Circuits Experiment I</b> , Emeritus Prof. Young-chul Chung  | 2023 |
| <b>Electronic Circuits Experiment II</b> , Emeritus Prof. Young-chul Chung | 2023 |
| <b>Electronic Circuits Experiment I</b> , Prof. Youngseok Choi             | 2022 |

## Technical Skills

**Programming Languages:** C, C++, Python, MATLAB  
**Frameworks & Libraries:** ROS, ROS2, TensorFlow, PyTorch  
**Simulation Tools:** MATLAB/Simulink, ns-3  
**Development Tools:** Linux, Git, Docker